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Greater China Semiconductors | Asia Pacific

Old Memory: Readacross from Kioxia IR Day

Legacy NAND technology powers high performance SSD: On its IR Day ([link](#)), Kioxia (covered by Kazuo Yoshikawa) highlighted its GP series high-performance SSD, which features 100m+ IOPS, ultra-fast and low-latency performance to accelerate next-gen AI. The company said it is compatible with Nvidia’s Storage-Next solution. The GP series SSD is used to expand GPU memory to resolve system memory constraints, and enables large-scale DB (database) and faster generation. The GP series SSD (XL-Flash) has adopted Kioxia’s proprietary SLC chip.

Implications for Greater China memory stocks: We mentioned in our note, [Old Memory: Upside Surprise Ahead](#), that legacy NAND technology could be ideal for fast reading and writing speeds and hence could see expanded TAM from adoption in datacenters. We view the confirmation from Kioxia as positive for GigaDevice, Winbond, and Macronix, which are legacy NAND vendors in our coverage and remain OW on all three stocks.

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific

Industry View

Attractive

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Valuation Methodology and Risks

Macronix International Co Ltd (2337.TW)

Base case, P/B, in line with our Greater China memory coverage, due to the memory industry's cyclical nature. We apply a target multiple of 7.0x to our 2026 BVPS estimate, vs. its historical average of 1.5x since 2017. We continue to believe the P/B target is justifiable, as The MLC and legacy TLC NAND could be in greater shortage into 2H26, with under supply easily going up to 40%. The only supplier which could fill the gap remains Macronix.

Risks to Upside

- Wider legacy NAND supply / demand gap
- NOR up-cycle, driven by limited supply expansion and/or stronger demand
- Faster growth for Switch hardware and core titles

Risks to Downside

- Less severe legacy NAND supply / demand gap
- NOR down-cycle, driven by faster supply expansion and/or demand weakness
- Slower growth for Switch hardware and core titles

Winbond Electronics Corp (2344.TW)

Base case, 5.5x 2026e P/B. This method is in line with the approach we adopt for Greater China memory IDM peers, in view of the industry's high volatility. We believe the stock merits a multiple beyond the high end of its historical band of 0.5-1.5x since 2017, as we are seeing DRAM pricing upside, a sustainable NOR price hike, and opportunities in SLC NAND and SiCap foundry business.

Risks to Upside

- NOR Flash pricing increase, driven by stronger-than-expected demand
- Stronger-than-expected DRAM pricing given ongoing supply shortage
- Faster-than-expected SLC NAND development

Risks to Downside

- NOR Flash downcycle, driven by weaker demand
- Lower-than-expected DRAM pricing given ongoing oversupply
- Slower-than-expected SLC NAND development

GigaDevice Semiconductor Beijing Inc (603986.SS)

Our price target is our base case value, derived from our residual income model. Key assumptions: cost of equity of 8.9% (beta 1.1, risk-free rate 2% and risk premium 6%), a payout ratio of 40%, medium-term growth rate of 16.4%, and a terminal growth rate of 3.0%.

Risks to Upside

- NOR up-cycle driven by stronger demand
- Superior chip design, leading to increasing exposure to low-density NOR
- Faster-than-expected DRAM development

Risks to Downside

- NOR down-cycle driven by weaker demand
- Inferior chip design, leading to increasing exposure to mid-/high-density NOR
- Slower-than-expected DRAM development

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(as of May 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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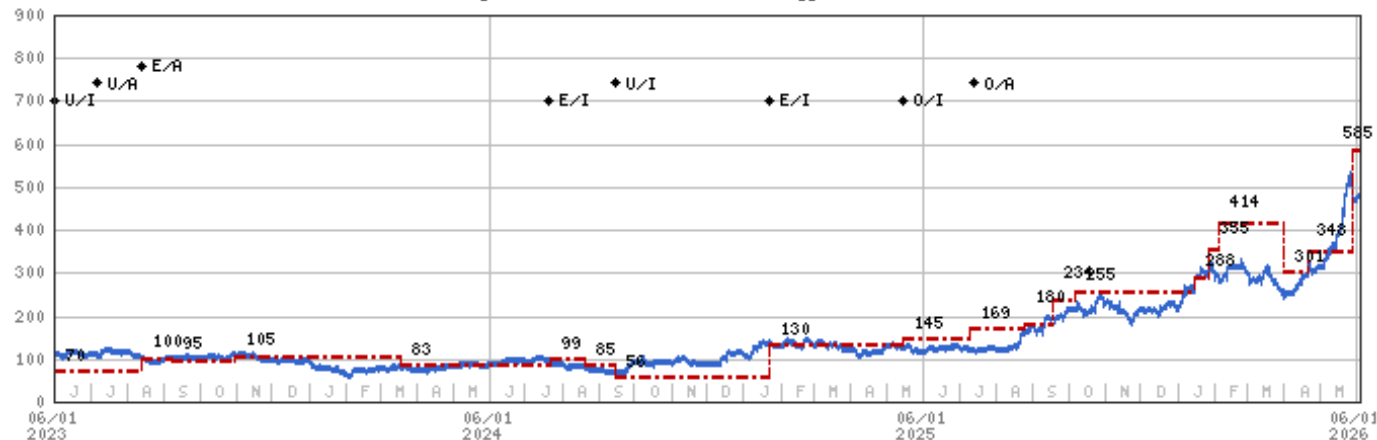
Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

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Stock Price, Price Target and Rating History (See Rating Definitions)

GigaDevice Semiconductor Beijing Inc (603986.SS) - As of 06/03/26 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 6/1/21 : O/I; 10/12/21 : O/C; 10/19/21 : E/C; 12/3/21 : O/C; 7/13/22 : E/C; 10/4/22 : E/A; 10/18/22 : U/A; 2/22/23 : U/I; 7/7/23 : U/A; 8/14/23 : E/A; 7/21/24 : E/I; 9/15/24 : U/I; 1/22/25 : E/I; 5/15/25 : O/I; 7/14/25 : O/A

Price Target History: 5/27/21 : 175; 7/19/21 : 222; 8/12/21 : 233; 9/17/21 : 208; 10/19/21 : 150; 12/3/21 : 222; 4/29/22 : 192; 7/5/22 : 168; 7/13/22 : 120; 8/27/22 : 122; 9/28/22 : 105; 10/18/22 : 72; 5/2/23 : 70; 8/14/23 : 100; 9/8/23 : 95; 10/31/23 : 105; 3/18/24 : 83; 7/23/24 : 99; 8/21/24 : 85; 9/15/24 : 56; 1/22/25 : 130; 5/15/25 : 145; 7/11/25 : 169; 8/25/25 : 180; 9/18/25 : 234; 10/7/25 : 255; 1/16/26 : 288; 1/28/26 : 355; 2/5/26 : 414; 3/31/26 : 301; 4/20/26 : 348; 5/28/26 : 585

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

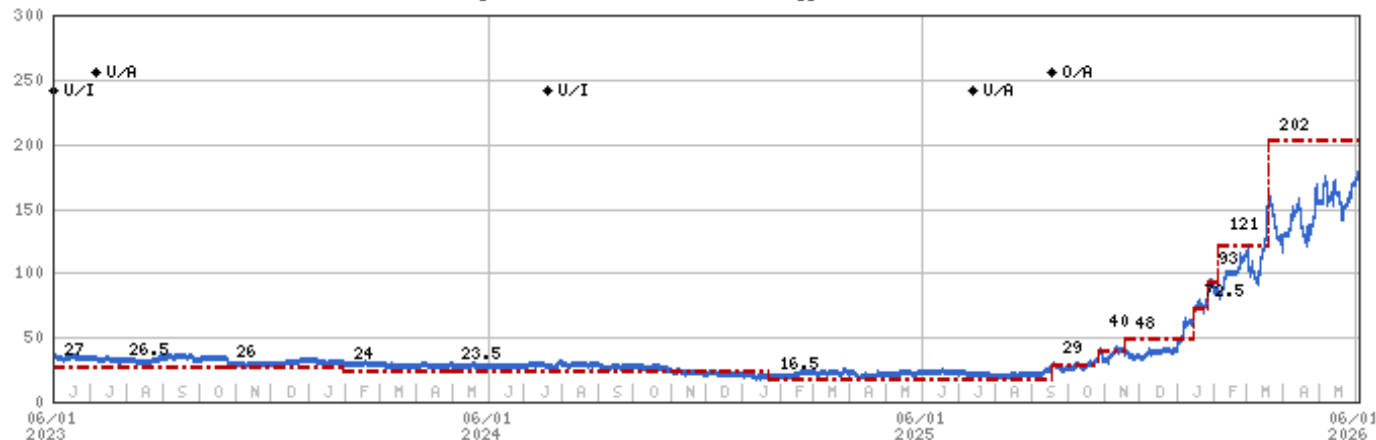
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Macronix International Co Ltd (2337.TW) - As of 06/03/26 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 6/1/21 : O/I; 8/12/21 : E/I; 10/12/21 : E/C; 10/19/21 : U/C; 10/4/22 : U/A; 2/22/23 : U/I; 7/7/23 : U/A;
7/21/24 : U/I; 7/14/25 : U/A; 9/18/25 : O/A

Price Target History: 5/18/21 : 47.5; 6/1/21 : 54; 8/12/21 : 44; 9/17/21 : 39; 10/19/21 : 30; 1/25/22 : 33; 7/5/22 : 27;
7/26/23 : 26.5; 10/24/23 : 26; 1/31/24 : 24; 5/1/24 : 23.5; 1/22/25 : 16.5; 9/18/25 : 29; 10/27/25 : 40; 11/18/25 : 48;
1/16/26 : 72.5; 1/28/26 : 93; 2/5/26 : 121; 3/20/26 : 202

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

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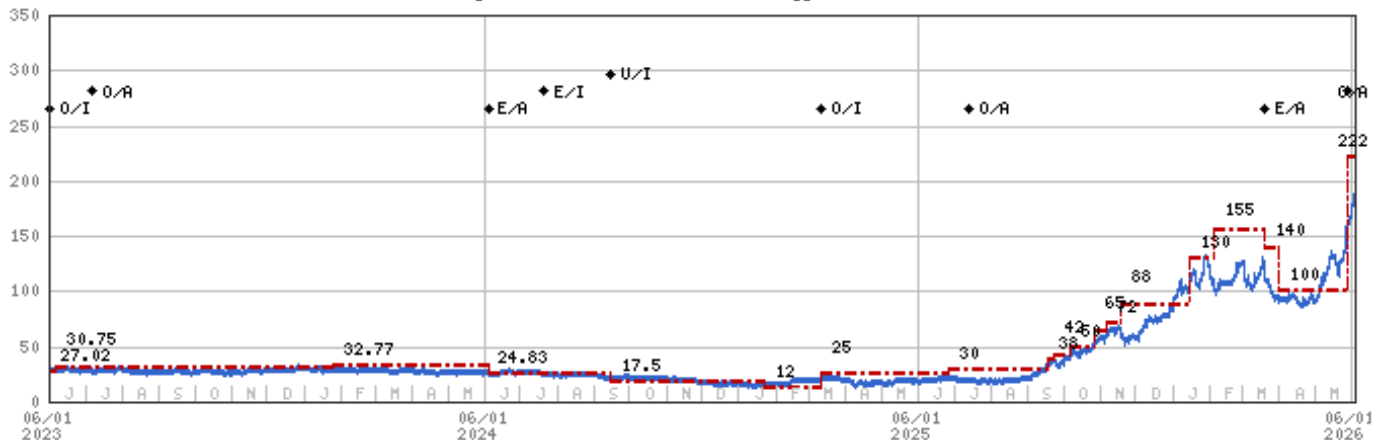
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Winbond Electronics Corp (2344.TW) - As of 06/03/26 GMT in TWD Industry : Greater China Technology Semiconductors



Stock Rating History: 6/1/21 : O/I; 8/12/21 : E/I; 10/12/21 : E/C; 10/19/21 : U/C; 10/4/22 : U/A; 2/22/23 : U/I; 3/2/23 : O/I; 7/7/23 : O/A; 6/4/24 : E/A; 7/21/24 : E/I; 9/15/24 : U/I; 3/11/25 : O/I; 7/14/25 : O/A; 3/20/26 : E/A; 5/28/26 : O/A
 Price Target History: 5/18/21 : 34.38; 6/1/21 : 39.3; 8/12/21 : 32.42; 9/17/21 : 28.98; 10/19/21 : 20.63; 2/21/22 : 26.52; 5/31/22 : 24.07; 7/5/22 : 16.7; 3/2/23 : 27.02; 6/6/23 : 30.75; 1/25/24 : 32.77; 6/4/24 : 24.83; 9/15/24 : 17.5; 1/22/25 : 12; 3/11/25 : 25; 6/26/25 : 30; 9/18/25 : 38; 9/23/25 : 42; 10/7/25 : 50; 10/27/25 : 65; 11/6/25 : 72; 11/18/25 : 88; 1/16/26 : 130; 2/5/26 : 155; 3/20/26 : 140; 3/31/26 : 100; 5/28/26 : 222

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
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 Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
 Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
 Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/02/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$91.92
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb282.80
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$168.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$4,575.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$590.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,300.00
Global Unichip Corp (3443.TW)	E (05/28/2026)	NT\$4,535.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$940.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$527.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$147.90
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$461.60
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$306.50
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb101.32
MediaTek (2454.TW)	O (11/28/2025)	NT\$4,525.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb663.47
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$419.50
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb602.80
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb92.65
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,820.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb106.58
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$578.00
SMIC (0981.HK)	O (10/21/2025)	HK\$81.95
TSMC (2330.TW)	O (02/07/2022)	NT\$2,380.00
UMC (2303.TW)	O (05/19/2026)	NT\$141.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$171.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$519.00

Daisy Dai, CFA

ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$184.80
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb67.65
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$167.50
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb105.30
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb33.03
Innoscence (2577.HK)	E (10/13/2025)	HK\$68.80
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb75.35
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$31.22
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb135.00
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb116.91
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb76.19
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb39.65
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb101.50

Daniel Yen, CFA

AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$1,075.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,590.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$18,095.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$122.50
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb162.72
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb474.69
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$175.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$380.20
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb237.00
Novatek (3034.TW)	U (02/04/2026)	NT\$507.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$211.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$827.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$85.90
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$653.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb55.64
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$184.50
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$119.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$290.50

Duan Liu

Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb126.50
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb519.50

Tiffany Yeh

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,010.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$804.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$23.98
Hon Precision (7769.TW)	O (04/17/2026)	NT\$7,175.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,515.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$301.27
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$8,080.00

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* Historical prices are not split adjusted.